



Physics

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The fundamental science of matter, energy, and the universe

Physics is the fundamental science studying matter, energy, and their interactions, seeking to understand the basic laws governing everything from subatomic particles to the entire universe.

● DEFINITION

The natural science that studies matter, its motion and behavior through space and time, and the related concepts of energy and force.

● WHY IT MATTERS

Physics provides the fundamental laws underlying all other physical sciences and engineering, from electricity to structural design to modern electronics.

● WORKING PRINCIPLE



Physicists develop mathematical theories to describe natural phenomena, testing predictions through carefully controlled experiments and refining theories based on observed results.

● QUICK FACTS

- ▶ Einstein's theory of relativity, developed over a century ago, is essential for modern GPS accuracy
- ▶ The Large Hadron Collider is the largest and most powerful particle accelerator ever built

● KEY TERMS

Mechanics Electromagnetism
Thermodynamics Quantum mechanics

● CORE CONCEPTS

- ▶ Classical mechanics
- ▶ Electromagnetism
- ▶ Thermodynamics
- ▶ Quantum mechanics

● APPLICATIONS

- ▶ Electronics and semiconductor design
- ▶ Medical imaging technology
- ▶ GPS satellite systems
- ▶ Renewable energy technology

● ADVANTAGES

- ▶ Foundation for nearly all other sciences and engineering
- ▶ Drives major technological breakthroughs
- ▶ Develops powerful analytical problem-solving skills

CAREER OPPORTUNITIES

Physicist, Research scientist, Data scientist, Engineering roles across many fields

RESEARCH AREAS

Quantum physics, Astrophysics, Particle physics

● REAL-LIFE EXAMPLES

- ▶ GPS relying on relativity corrections
- ▶ MRI machines using nuclear magnetic principles
- ▶ Solar panels using the photoelectric effect

● LIMITATIONS

- ▶ Advanced theoretical physics can require years of specialized study
- ▶ Some phenomena remain unexplained by current theories
- ▶ Experimental physics often requires expensive equipment

INDUSTRY APPLICATIONS

Technology, Energy, Aerospace, Healthcare (medical physics)

FUTURE SCOPE

Ongoing research into quantum computing, dark matter, and fusion energy represents some of physics' most active frontiers.

● CURRENT TRENDS

Growing convergence of physics with computer science in fields like quantum computing and AI-driven scientific discovery.

● SUMMARY

Physics studies the fundamental laws of matter and energy, forming the foundation for nearly all other scientific and engineering fields.

